

# Farthest pair of points (tag: far)

---

Training suite    Available memory: 512 MB. Maximum running time: ? s.

You are given a set of points on the plane. Your task is to find two points whose Euclidean distance is as large as possible.

## Input

The first line contains a single integer  $n$  ( $2 \leq n \leq 200\,000$ ), the number of points.

Each of the next  $n$  lines contains two integers  $x$  and  $y$ , denoting the coordinates of a point. The coordinates satisfy  $-10^9 \leq x, y \leq 10^9$ . No point appears more than once.

## Output

Output two lines, each containing the coordinates of one point from the input.

The two points should be chosen such that the Euclidean distance between them is maximized. If there are multiple valid answers, output any of them.

## Subtasks

| Subtask | Constraints               |
|---------|---------------------------|
| 1       | $n \leq 10\,000$          |
| 2       | No additional constraints |